



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE/ NATIONAL SENIOR CERTIFICATE

GRADE 12

MATHEMATICS P2

NOVEMBER 2020

MARKS: 150

TIME: 3 hours

**This question paper consists of 14 pages, 1 information sheet
and an answer book of 24 pages.**

INSTRUCTIONS AND INFORMATION

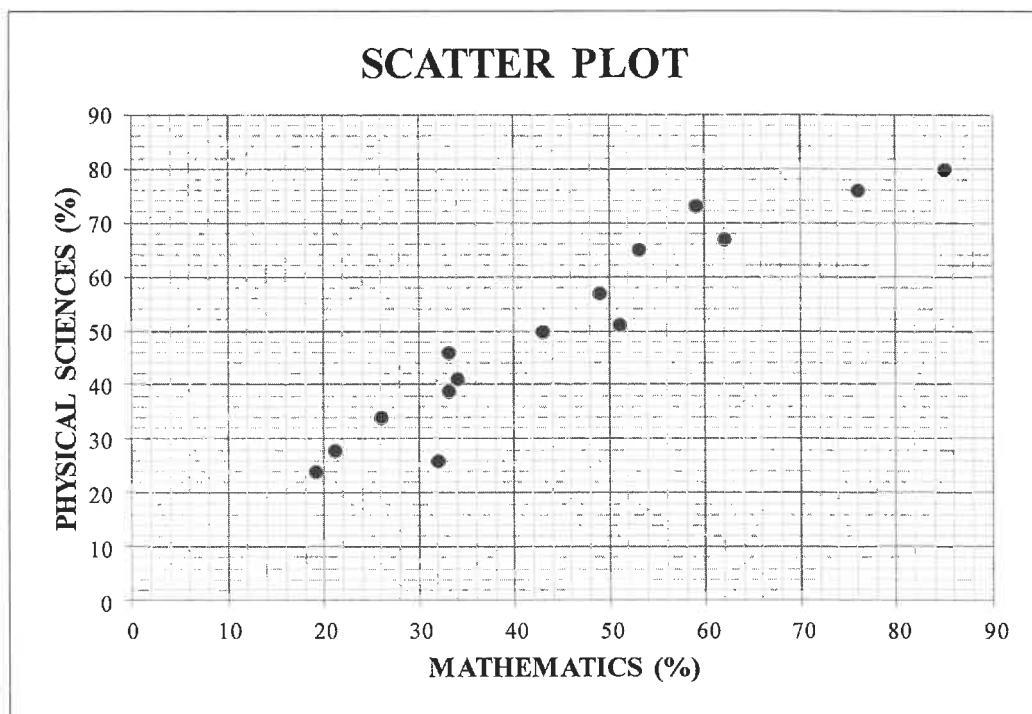
Read the following instructions carefully before answering the questions.

1. This question paper consists of 10 questions.
2. Answer ALL the questions in the SPECIAL ANSWER BOOK provided.
3. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
4. Answers only will NOT necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
6. If necessary, round off answers correct to TWO decimal places, unless stated otherwise.
7. Diagrams are NOT necessarily drawn to scale.
8. An information sheet with formulae is included at the end of the question paper.
9. Write neatly and legibly.

QUESTION 1

A Mathematics teacher was curious to establish if her learners' Mathematics marks influenced their Physical Sciences marks. In the table below, the Mathematics and Physical Sciences marks of 15 learners in her class are given as percentages (%).

MATHEMATICS (AS %)	26	62	21	33	53	76	32	59	43	33	49	51	19	34	85
PHYSICAL SCIENCES (AS %)	34	67	28	46	65	76	26	73	50	39	57	51	24	41	80

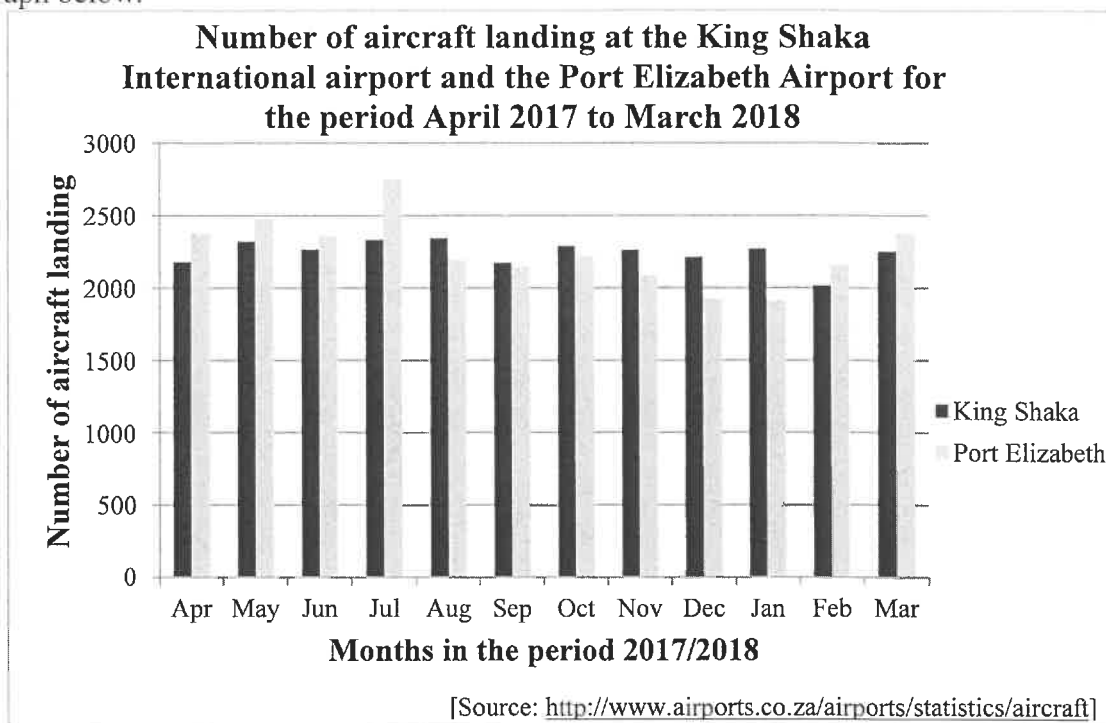


- 1.1 Determine the equation of the least squares regression line for the data. (3)
- 1.2 Draw the least squares regression line on the scatter plot provided in the ANSWER BOOK. (2)
- 1.3 Predict the Physical Sciences mark of a learner who achieved 69% for Mathematics. (2)
- 1.4 Write down the correlation coefficient between the Mathematics and Physical Sciences marks for the data. (1)
- 1.5 Comment on the strength of the correlation between the Mathematics and Physical Sciences marks for the data. (1)
- 1.6 What trend did the teacher observe between the results of the two subjects? (1)

[10]

QUESTION 2

The number of aircraft landing at the King Shaka International Airport and the Port Elizabeth Airport for the period starting in April 2017 and ending in March 2018, is shown in the double bar graph below.



- 2.1 The number of aircraft landing at the Port Elizabeth Airport exceeds the number of aircraft landing at the King Shaka International Airport during some months of the given period. During which month is this difference the greatest? (1)

- 2.2 The number of aircraft landing at the King Shaka International Airport during these months are:

2 182	2 323	2 267	2 334	2 346	2 175
2 293	2 263	2 215	2 271	2 018	2 254

- Calculate the mean for the data. (2)

- 2.3 Calculate the standard deviation for the number of aircraft landing at the King Shaka International Airport for the given period. (2)

- 2.4 Determine the number of months in which the number of aircraft landing at the King Shaka International Airport were within one standard deviation of the mean. (3)

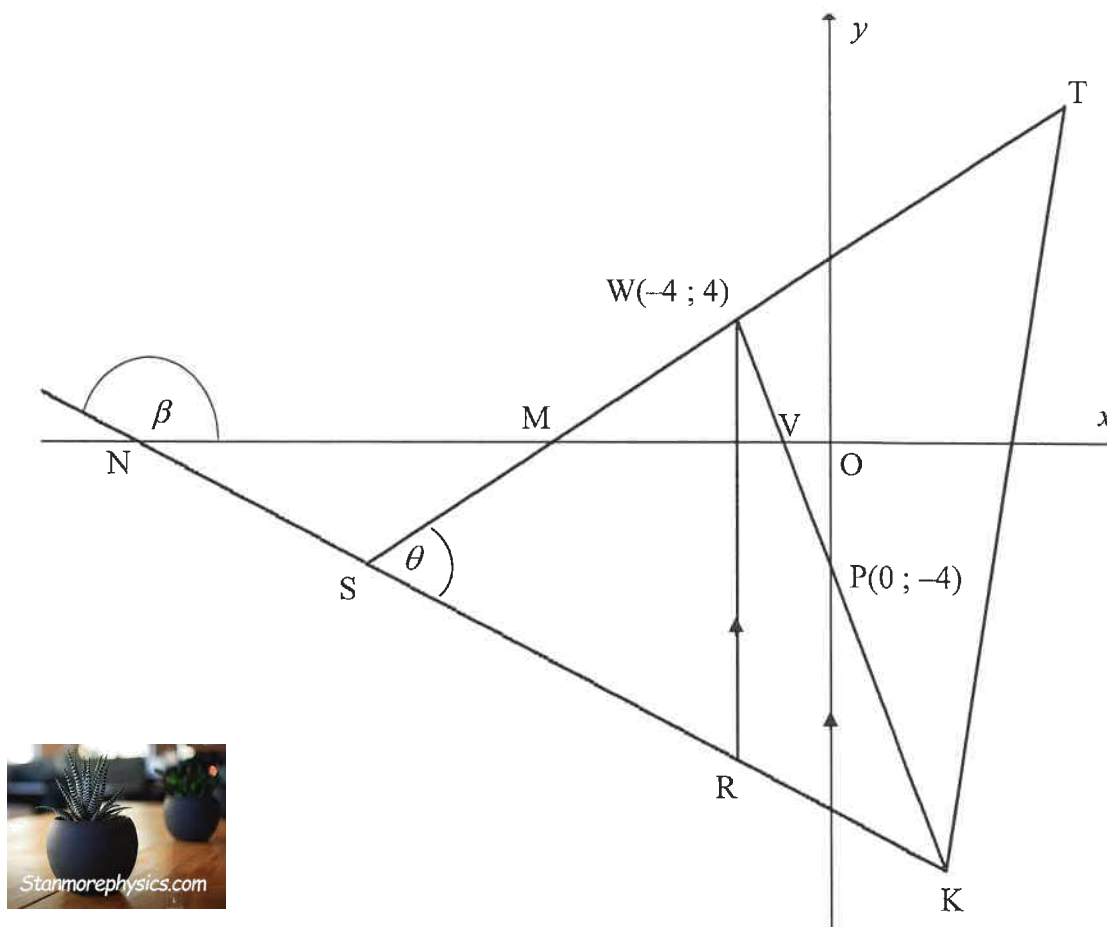
- 2.5 Which ONE of the following statements is CORRECT?

- A. During December and January, there were more landings at the Port Elizabeth Airport than at the King Shaka International Airport.
- B. There was a greater variation in the number of aircraft landing at the King Shaka International Airport than at the Port Elizabeth Airport for the given period.
- C. The standard deviation of the number of landings at the Port Elizabeth Airport will be higher than the standard deviation of the number of landings at the King Shaka International Airport.

(1)
[9]

QUESTION 3

$\triangle TSK$ is drawn. The equation of ST is $y = \frac{1}{2}x + 6$ and ST cuts the x -axis at M . $W(-4; 4)$ lies on ST and R lies on SK such that WR is parallel to the y -axis. WK cuts the x -axis at V and the y -axis at $P(0; -4)$. KS produced cuts the x -axis at N . $\hat{T}SK = \theta$.

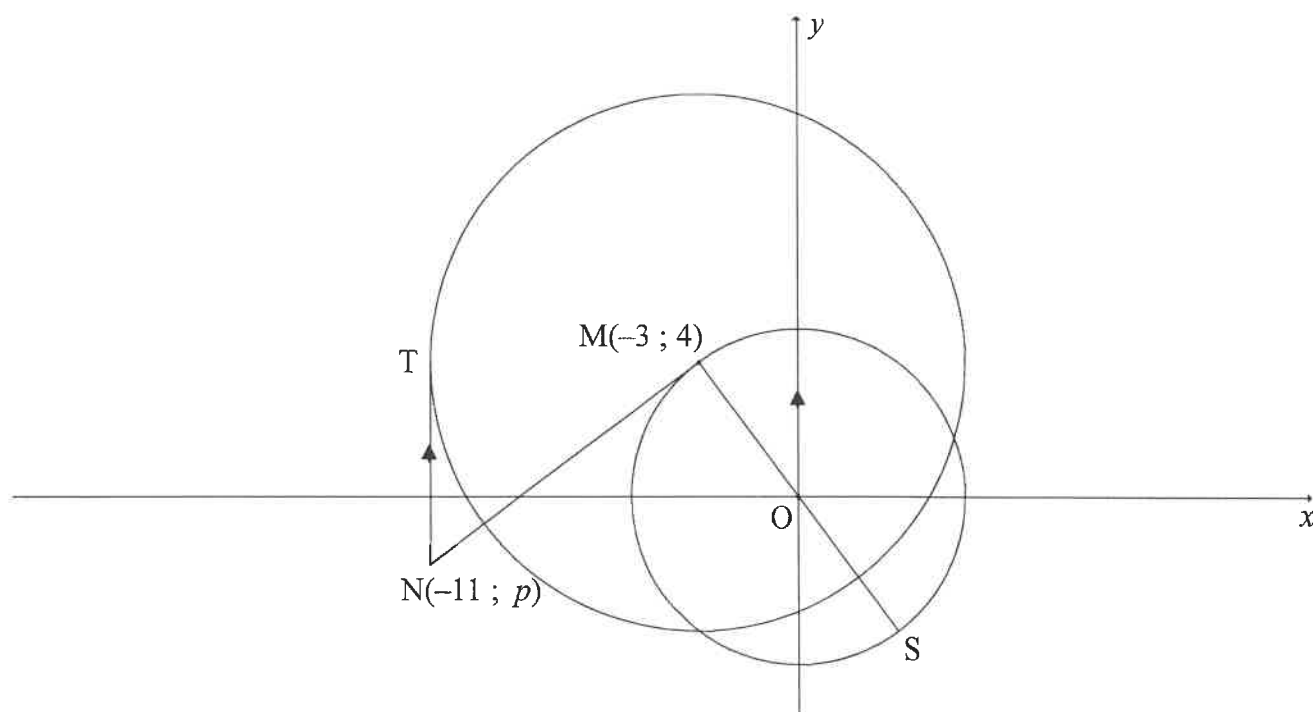


- 3.1 Calculate the gradient of WP . (2)
- 3.2 Show that $WP \perp ST$. (2)
- 3.3 If the equation of SK is given as $5y + 2x + 60 = 0$, calculate the coordinates of S . (4)
- 3.4 Calculate the length of WR . (4)
- 3.5 Calculate the size of θ . (5)
- 3.6 Let L be a point in the third quadrant such that $SWRL$, in that order, forms a parallelogram. Calculate the area of $SWRL$. (4)

[21]

QUESTION 4

$M(-3 ; 4)$ is the centre of the large circle and a point on the small circle having centre $O(0 ; 0)$. From $N(-11 ; p)$, a tangent is drawn to touch the large circle at T with NT is parallel to the y -axis. NM is a tangent to the smaller circle at M with MOS a diameter.

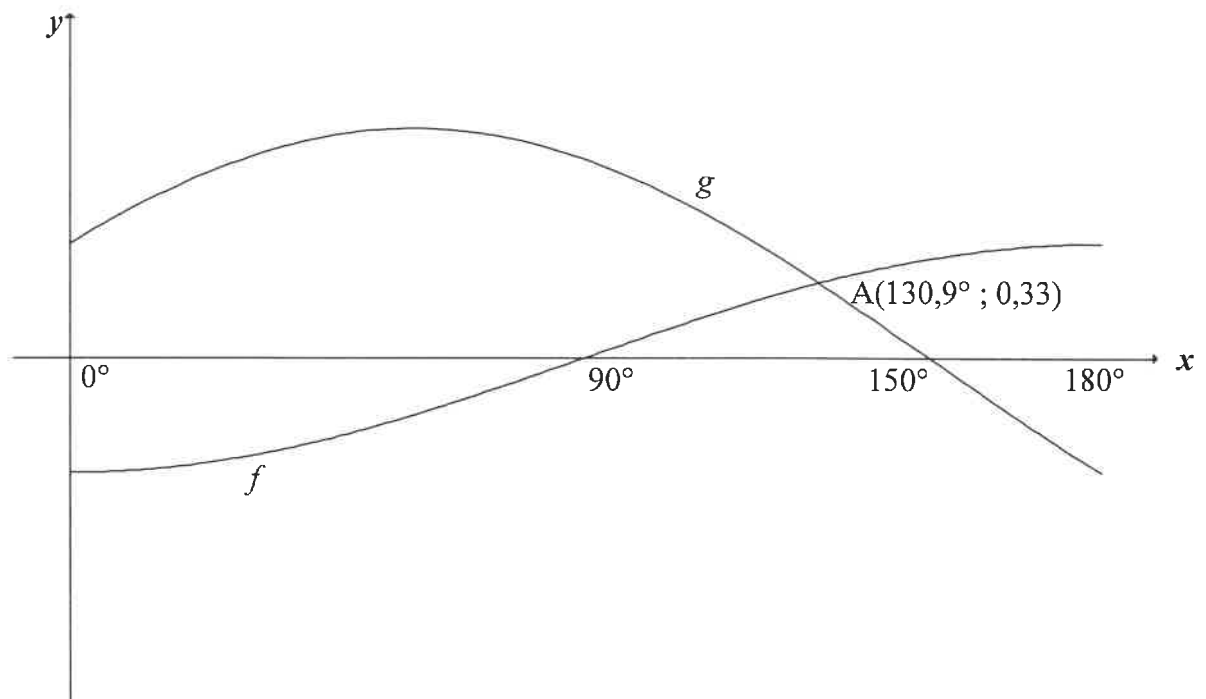


- 4.1 Determine the equation of the small circle. (2)
- 4.2 Determine the equation of the circle centred at M in the form $(x - a)^2 + (y - b)^2 = r^2$ (3)
- 4.3 Determine the equation of NM in the form $y = mx + c$ (4)
- 4.4 Calculate the length of SN . (5)
- 4.5 If another circle with centre $B(-2 ; 5)$ and radius k touches the circle centred at M , determine the value(s) of k , correct to ONE decimal place. (5)

[19]

QUESTION 5

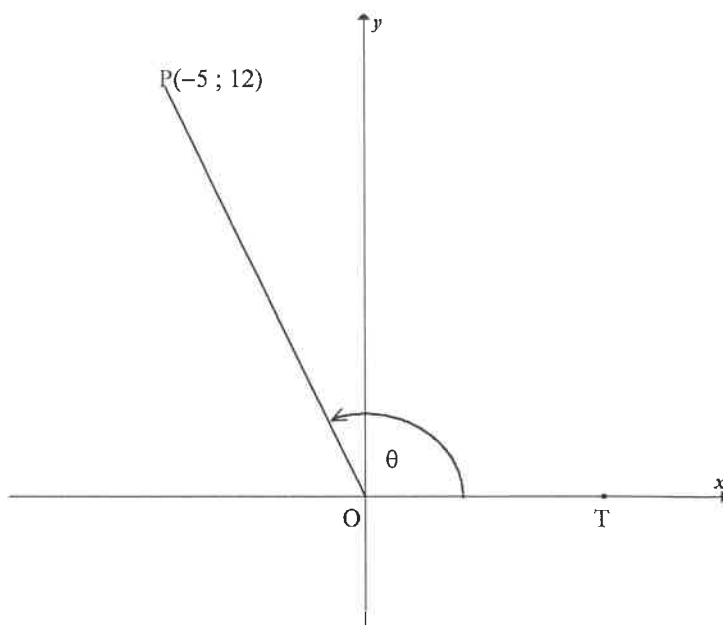
The graphs of $f(x) = -\frac{1}{2}\cos x$ and $g(x) = \sin(x+30^\circ)$, for the interval $x \in [0^\circ; 180^\circ]$, are drawn below. $A(130,9^\circ; 0,33)$ is the approximate point of intersection of the two graphs.



- 5.1 Write down the period of g . (1)
- 5.2 Write down the amplitude of f . (1)
- 5.3 Determine the value of $f(180^\circ) - g(180^\circ)$. (1)
- 5.4 Use the graphs to determine the values of x , in the interval $x \in [0^\circ; 180^\circ]$, for which:
- 5.4.1 $f(x-10^\circ) = g(x-10^\circ)$ (1)
- 5.4.2 $\sqrt{3}\sin x + \cos x \geq 1$ (4)
- [8]**

QUESTION 6

- 6.1 In the diagram, $P(-5 ; 12)$ and T lies on the positive x -axis. $\widehat{POT} = \theta$



Answer the following **without using a calculator**:

- 6.1.1 Write down the value of $\tan \theta$ (1)
- 6.1.2 Calculate the value of $\cos \theta$ (3)
- 6.1.3 $S(a ; b)$ is a point in the third quadrant such that $\widehat{TOS} = \theta + 90^\circ$ and $OS = 6,5$ units. Calculate the value of b . (4)
- 6.2 Determine, **without using a calculator**, the value of the following trigonometric expression:
- $$\frac{\sin 2x \cdot \cos(-x) + \cos 2x \cdot \sin(360^\circ - x)}{\sin(180^\circ + x)} \quad (5)$$
- 6.3 Determine the general solution of the following equation:
- $$6 \sin^2 x + 7 \cos x - 3 = 0 \quad (6)$$
- 6.4 Given: $x + \frac{1}{x} = 3 \cos A$ and $x^2 + \frac{1}{x^2} = 2$

Determine the value of $\cos 2A$ **without using a calculator**.

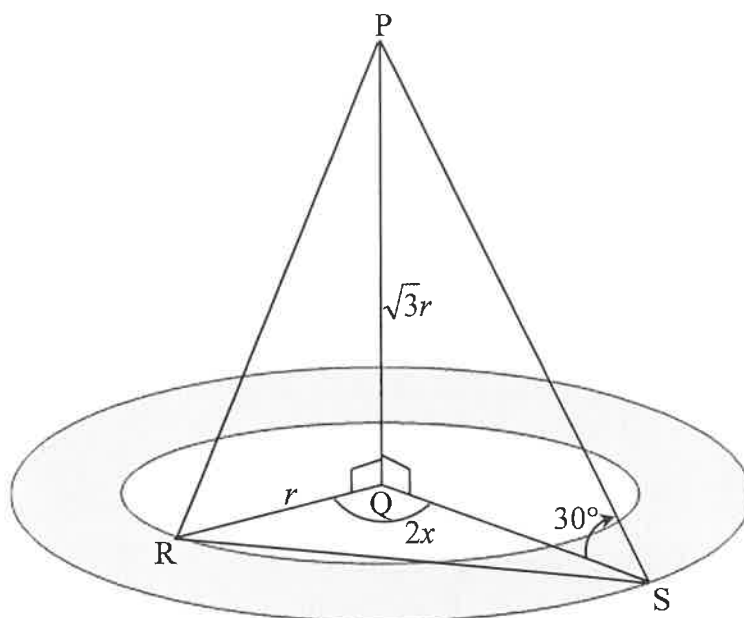
(5)

[24]

QUESTION 7

A landscape artist plans to plant flowers within two concentric circles around a vertical light pole PQ. R is a point on the inner circle and S is a point on the outer circle. R, Q and S lie in the same horizontal plane. RS is a pipe used for the irrigation system in the garden.

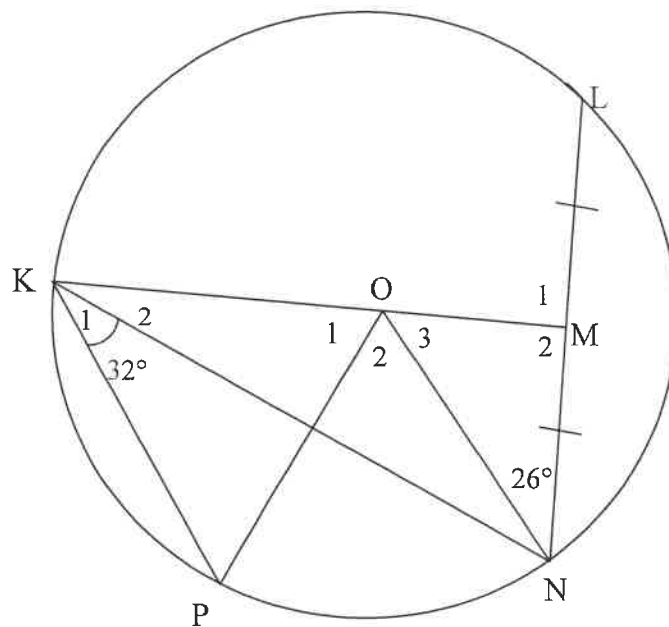
- The radius of the inner circle is r units and the radius of the outer circle is QS .
- The angle of elevation from S to P is 30° .
- $\angle RQS = 2x$ and $PQ = \sqrt{3}r$



- 7.1 Show that $QS = 3r$ (3)
- 7.2 Determine, in terms of r , the area of the flower garden. (2)
- 7.3 Show that $RS = r\sqrt{10 - 6 \cos 2x}$ (3)
- 7.4 If $r = 10$ metres and $x = 56^\circ$, calculate RS. (2)
- [10]

QUESTION 8

- 8.1 O is the centre of the circle.. KOM bisects chord LN and $\hat{MNO} = 26^\circ$. K and P are points on the circle with $\hat{NKP} = 32^\circ$. OP is drawn.

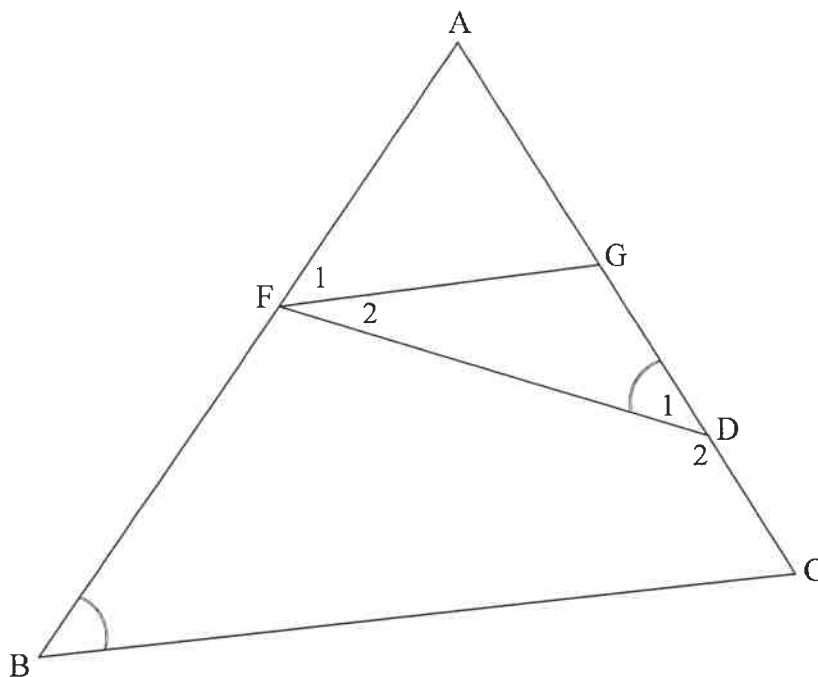


- 8.1.1 Determine, giving reasons, the size of:

- (a) $\hat{O}_2$ (2)
- (b) $\hat{O}_1$ (4)

- 8.1.2 Prove, giving reasons, that KN bisects $\hat{OKP}$. (3)

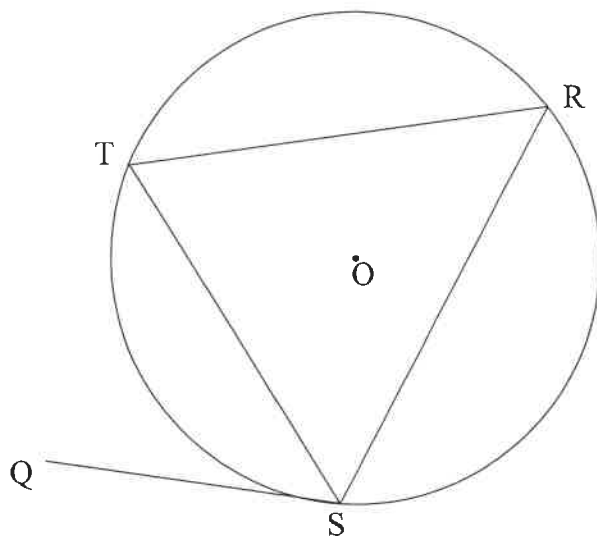
- 8.2 In $\triangle ABC$, F and G are points on sides AB and AC respectively. D is a point on GC such that $\hat{D}_1 = \hat{B}$.



- 8.2.1 If AF is a tangent to the circle passing through points F, G and D, then prove, giving reasons, that $FG \parallel BC$. (4)
- 8.2.2 If it is further given that $\frac{AF}{FB} = \frac{2}{5}$, $AC = 2x - 6$ and $GC = x + 9$, then calculate the value of x . (4)
- [17]

QUESTION 9

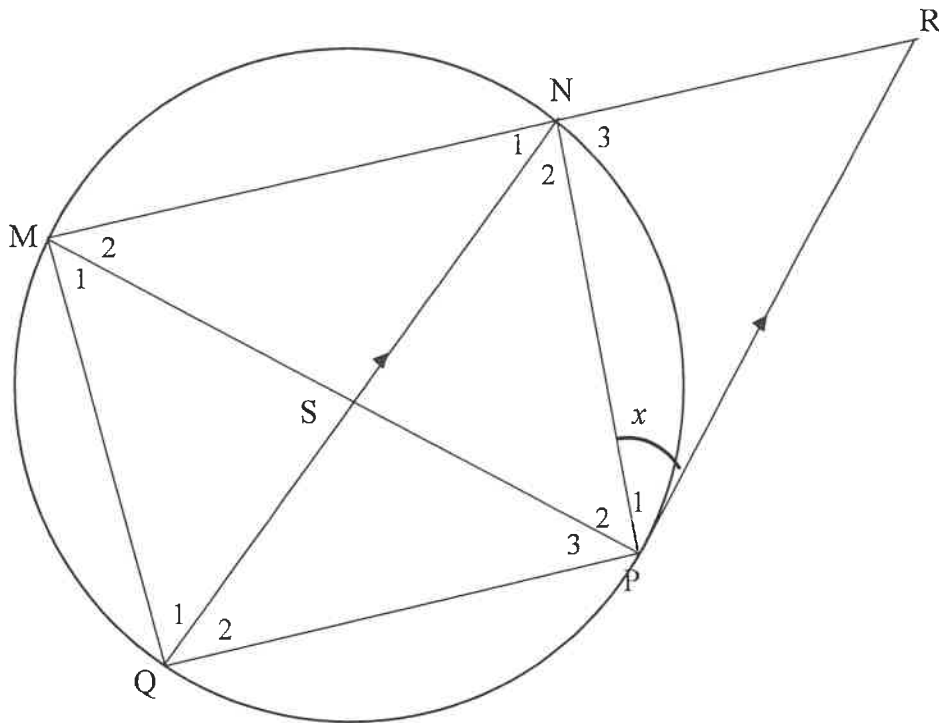
- 9.1 In the diagram, O is the centre of the circle. Points S, T and R lie on the circle. Chords ST, SR and TR are drawn in the circle. QS is a tangent to the circle at S.



Use the diagram to prove the theorem which states that $\hat{QST} = \hat{R}$.

(5)

- 9.2 Chord QN bisects $\widehat{MNP}$ and intersects chord MP at S. The tangent at P meets MN produced at R such that $QN \parallel PR$. Let $\hat{P}_1 = x$.



- 9.2.1 Determine the following angles in terms of x . Give reasons

(a) $\hat{N}_2$ (2)

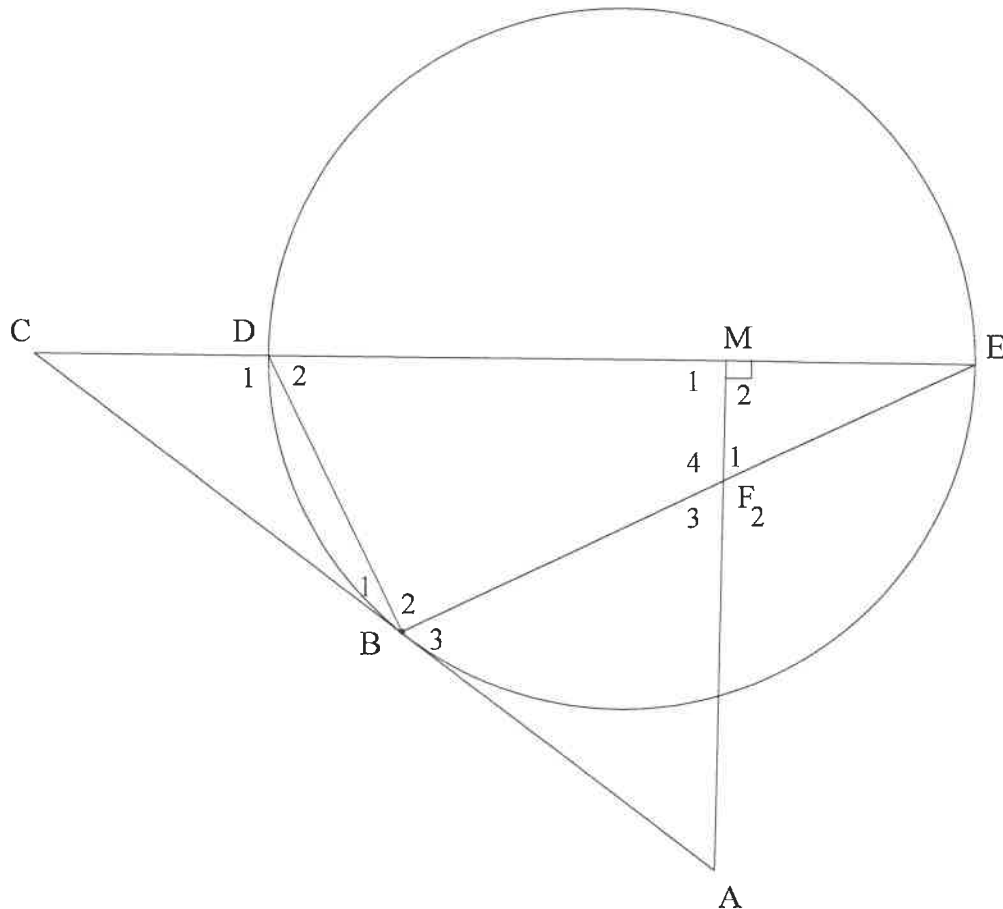
(b) $\hat{Q}_2$ (2)

- 9.2.2 Prove, giving reasons, that $\frac{MN}{NR} = \frac{MS}{SQ}$ (6)

[15]

QUESTION 10

In the diagram, a circle passes through D, B and E. Diameter ED of the circle is produced to C and AC is a tangent to the circle at B. M is a point on DE such that $AM \perp DE$. AM and chord BE intersect at F. AM and chord BE intersect at F.



10.1 Prove, giving reasons, that:

10.1.1 FBDM is a cyclic quadrilateral (3)

10.1.2 $\hat{B}_3 = \hat{F}_1$ (4)

10.1.3 $\triangle CDB \parallel \triangle CBE$ (3)

10.2 If it is further given that $CD = 2$ units and $DE = 6$ units, calculate the length of:

10.2.1 BC (3)

10.2.2 DB (4)

[17]

TOTAL: 150

INFORMATION SHEET

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}; r \neq 1$$

$$S_\infty = \frac{a}{1 - r}; -1 < r < 1$$

$$F = \frac{x[(1 + i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x - a)^2 + (y - b)^2 = r^2$$

$$\text{In } \triangle ABC: \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2}ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2\sin^2 \alpha \\ 2\cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

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SC/NSC Answer Book
SS/NSS Antwoordeboek

CENTRE NUMBER
SENTRUMNOMMER

EXAMINATION NUMBER
EKSAMENNUMMER

DATE
DATUM

BOOK NUMBER
BOEKNOMMER

OF
VAN

BOOKS
ROEKE

SUBJECT CODE
VAKKODE

PAPER NUMBER
VRAESTELNOMMER

2

SUBJECT NAME
VAKNAAM

MATHEMATICS/WISKUNDE[illegible]

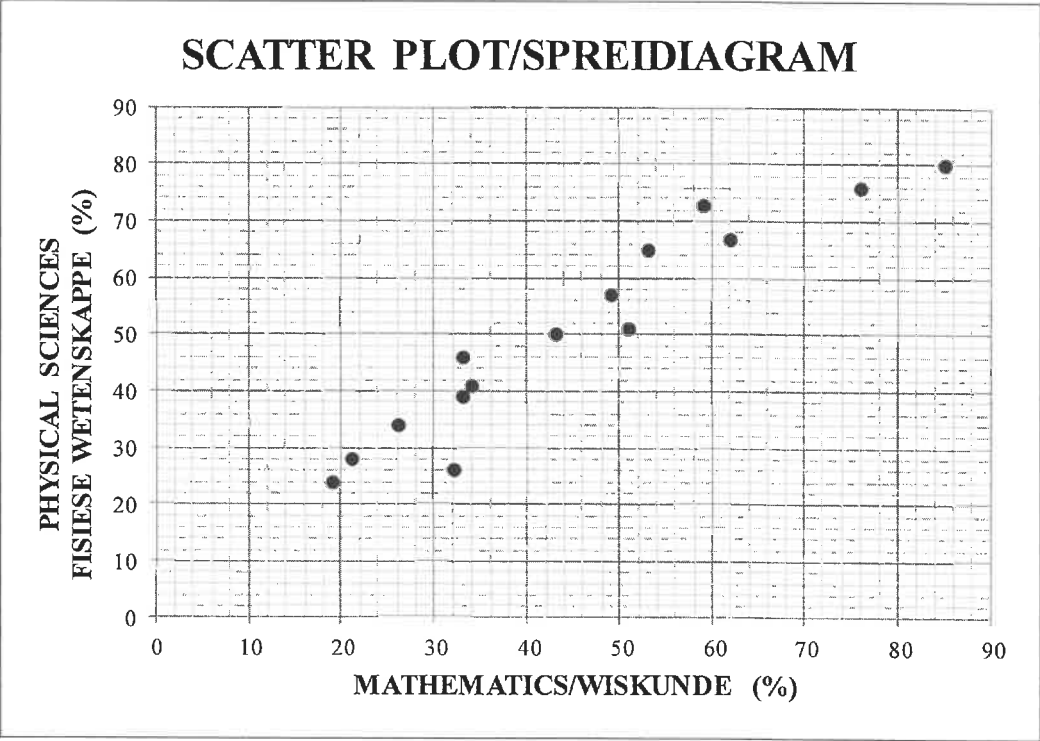
READ INSTRUCTIONS ON THE NEXT PAGE.
LEES INSTRUKSIES OP VOLGENDE BLADSY.

This answer book consists of 24 pages./Hierdie antwoordeboek bestaan uit 24 bladsye.

PLEASE FOLLOW THESE INSTRUCTIONS CAREFULLY	VOLG ASSEBLIEF HIERDIE INSTRUKSIES NOUKEURIG
- Clearly write your examination number and centre number in the space provided and attach your barcode label in the space provided. - Remember that your own name (or the name of your school) may NOT appear anywhere on or in this answer book. - Answer ALL questions in the spaces provided. - NO pages may be torn from this answer book. - Read the instructions printed on your timetable carefully as well as any other instructions which may be given in each question paper. - Candidates may NOT retain an answer book or remove it from the examination room. - Answers must be written in black/blue ink as distinctly as possible. Do NOT write in the margins. - Write the numbers of the questions you have answered on the front cover of the answer book where marks are to be recorded. - If you require additional space for your answers: - Use the additional space provided at the end of the answer book. - When answering a question in the additional space, clearly indicate the question number in the column on the LHS. - Rule off after each answer. - Draw a neat line through any work/rough work that must not be marked.	- Skryf jou eksamennommer en sentrumnummer duidelik in die ruimtes verskaf en plak jou stafieskodeplakker in die ruimte verskaf. - Onthou dat jou eie naam (of die naam van jou skool) NIE op of in hierdie antwoordeboek mag voorkom NIE. - Beantwoord ALLE vrae in die ruimtes wat verskaf is. - GEEN bladsye mag uit hierdie antwoordeboek geskeur word NIE. - Lees die instruksies, wat op jou eksamenrooster gedruk is, sorgvuldig deur, asook enige ander instruksies wat op elke vraestel gegee word. - GEEN antwoordeboek mag deur die kandidaat behou of uit die eksamenlokaal verwyder word NIE. - Skryf die antwoorde so duidelik moontlik met swart/blou ink. Laat die kantlyne oop. - Skryf die nommers van die vrae wat jy beantwoord het op die voorblad van die antwoordeboek waar die punte aangebring word. - In geval jy bykomende ruimte benodig vir jou antwoorde: - Gebruik die bykomende ruimte wat aan die einde van die antwoordeboek verskaf word. - As 'n vraag in die bykomende ruimte beantwoord word, dui duidelik die vraagnommer in die kolom aan die LK aan. - Trek 'n lyn na elke antwoord. - Trek 'n netjiese lyn deur enige werk/rofwerk wat nie nagesien moet word nie.

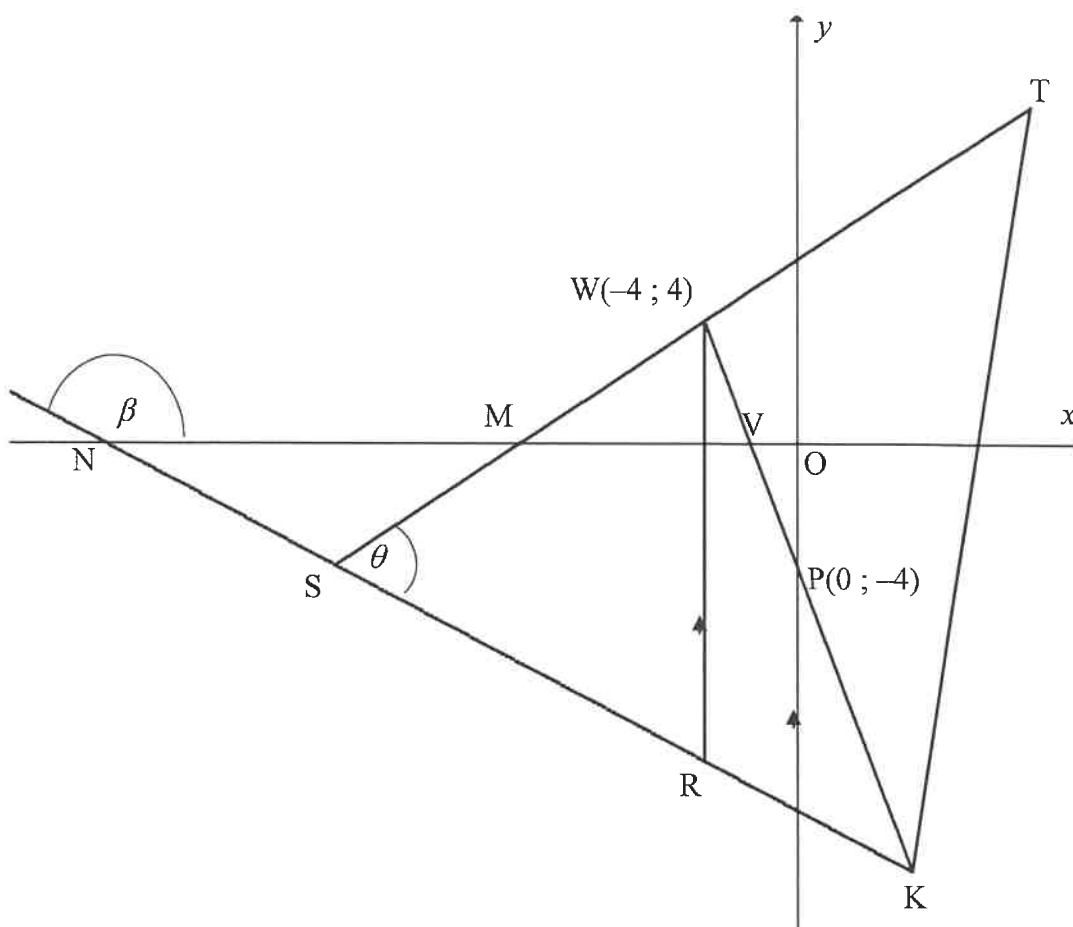
QUESTION/VRAAG 1

Mathematics <i>Wiskunde</i> (as %)	26	62	21	33	53	76	32	59	43	33	49	51	19	34	85
Physical Sciences <i>Fisiese Wetenskappe</i> (as %)	34	67	28	46	65	76	26	73	50	39	57	51	24	41	80

	Solution/Oplissing	Marks Punte
1.1		(3)
1.2	SCATTER PLOT/SPREIDIAGRAM  (2)	(2)
1.3		(2)

	Solution/Oplossing	Marks Punte
1.4		(1)
1.5		(1)
1.6		(1)
		[10]

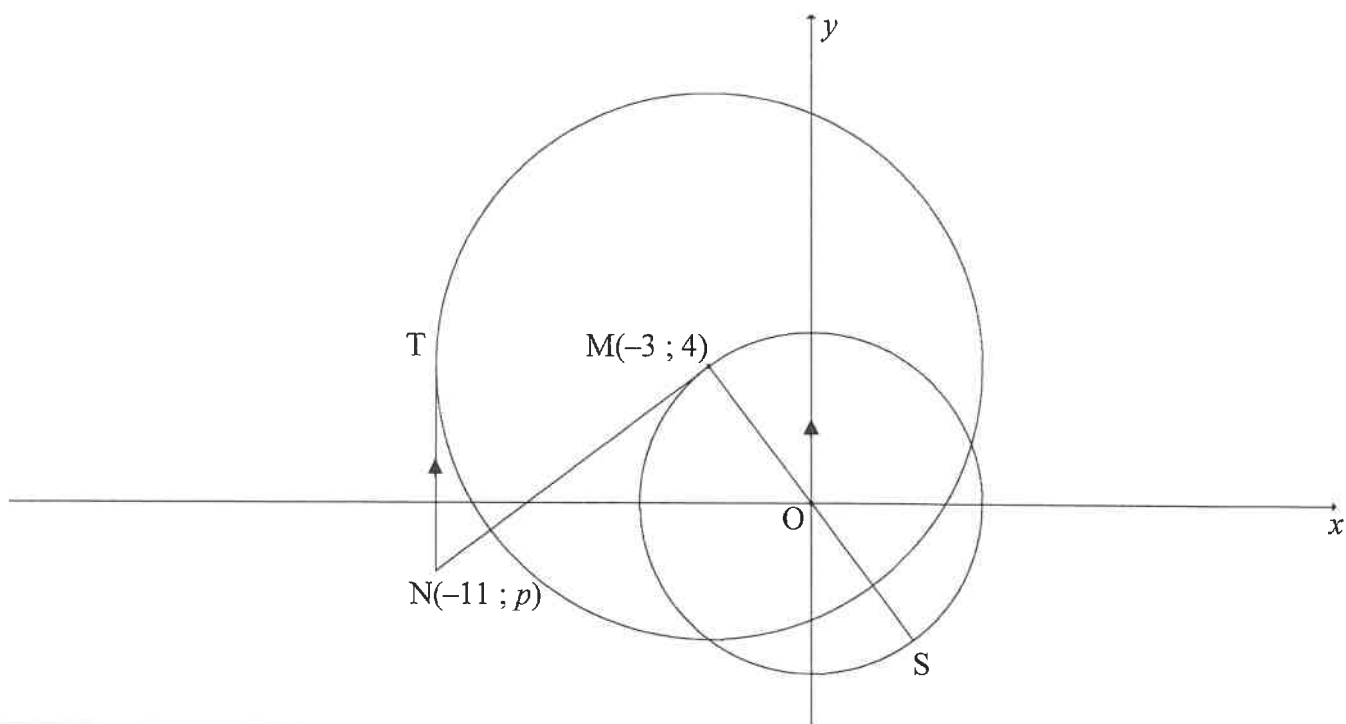
QUESTION/VRAAG 3



	Solution/Oplissing	Marks Punte
3.1		(2)
3.2		(2)
3.3		(4)

	Solution/Oplissing	Marks Punte
3.4		(4)
3.5		(5)
3.6		(4)
		[21]

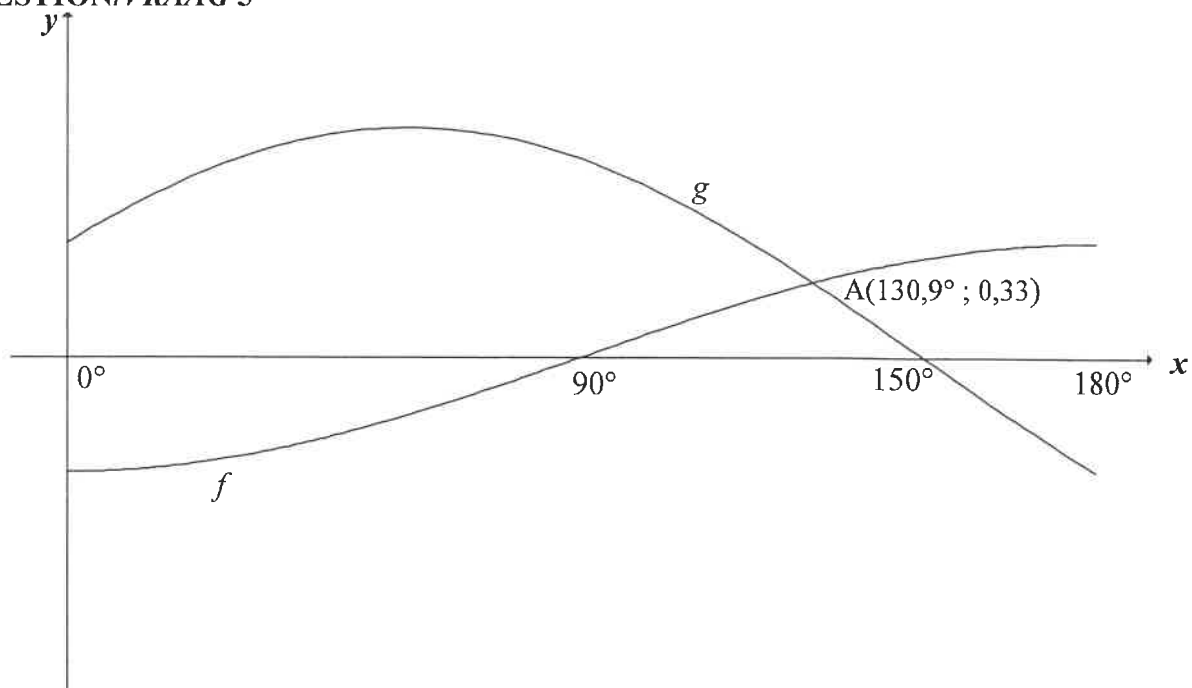
QUESTION/VRAAG 4



	Solution/Oplissing	Marks Punte
4.1		(2)
4.2		(3)
4.3		(4)

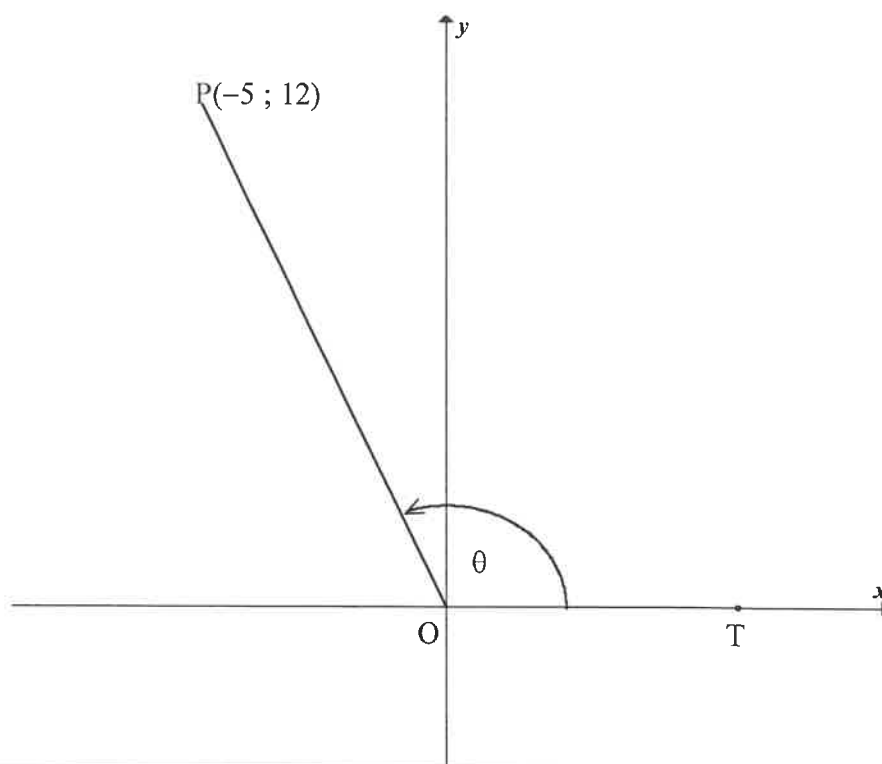
	Solution/<i>Oplossing</i>	Marks <i>Punte</i>
4.4		(5)
4.5		(5)
		[19]

QUESTION/VRAAG 5



	Solution/Oplissing	Marks Punte
5.1		(1)
5.2		(1)
5.3		(1)
5.4.1		(1)
5.4.2		(4)
		[8]

QUESTION/VRAAG 6

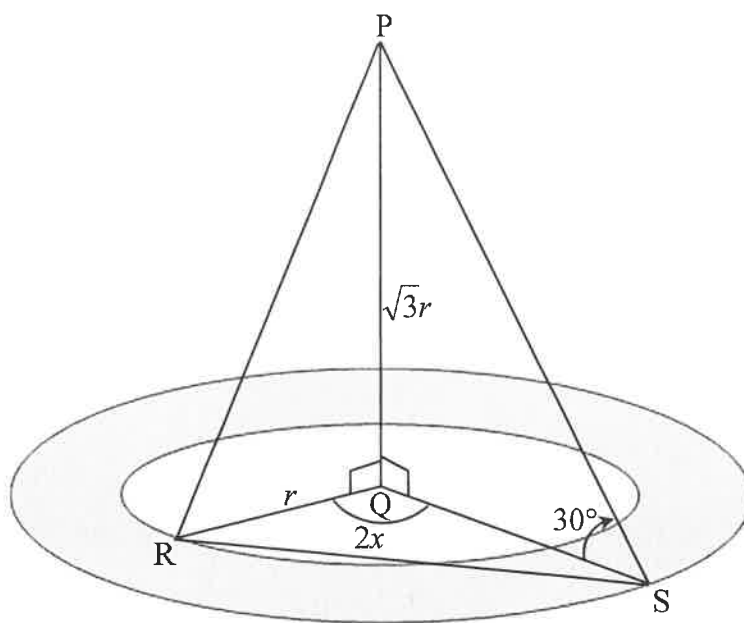


	Solution/Oplissing	Marks Punte
6.1.1		(1)
6.1.2		(3)
6.1.3		(4)

6.2		(5)
6.3		

6.4		(5)
		[24]

QUESTION/VRAAG 7



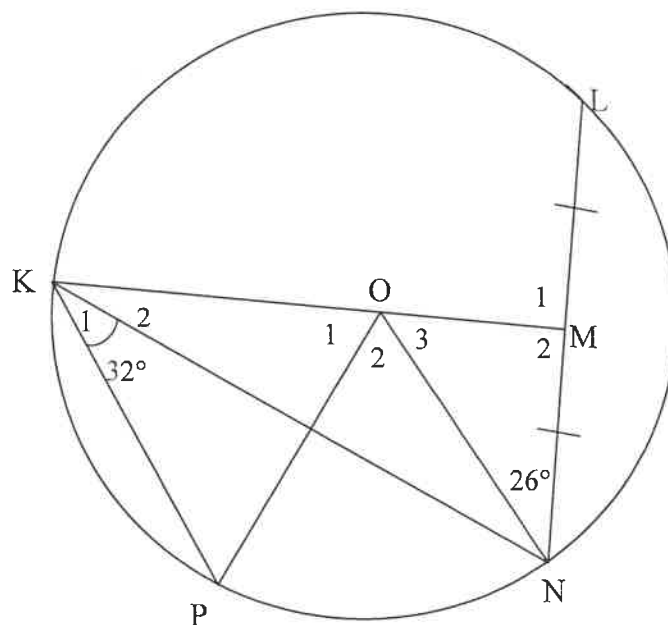
	Solution/Oplissing	Marks Punte
7.1		(3)
7.2		(2)
7.3		(3)

7.4		(2)
		[10]

Give reasons for your statements in QUESTIONS 8, 9 and 10.
 Verskaf redes vir jou bewerings in VRAAG 8, 9 en 10.

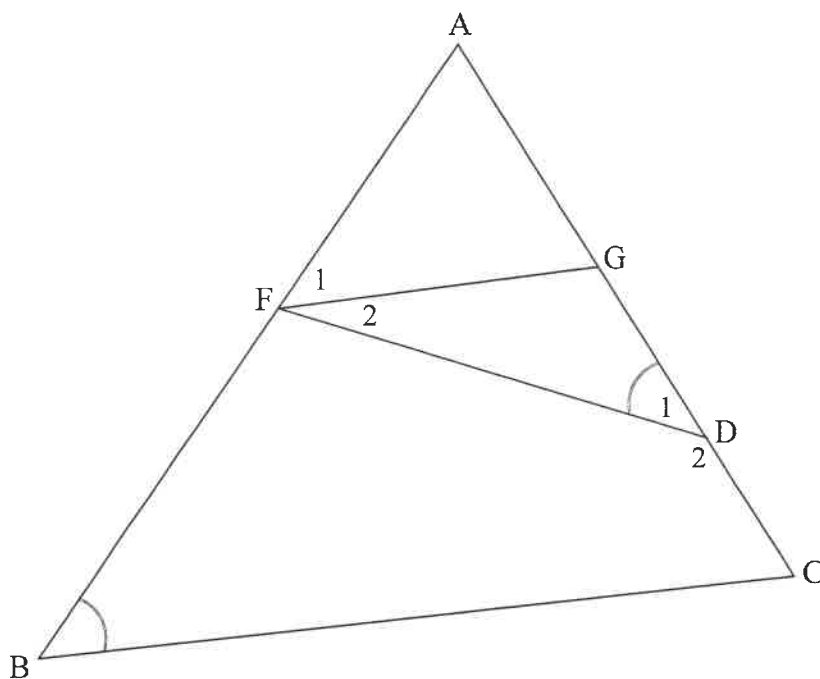
QUESTION/VRAAG 8

8.1



	Solution/Oplissing	Marks Punte
8.1.1(a)		(2)
8.1.1(b)		(4)
8.1.2		(3)

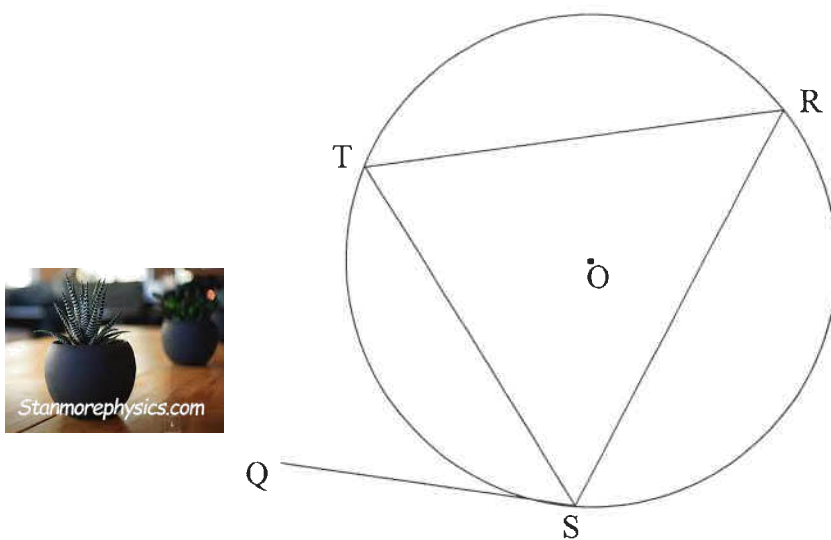
8.2



	Solution/Oplissing	Marks Punte
8.2.1		(4)
8.2.2		(4)
		[17]

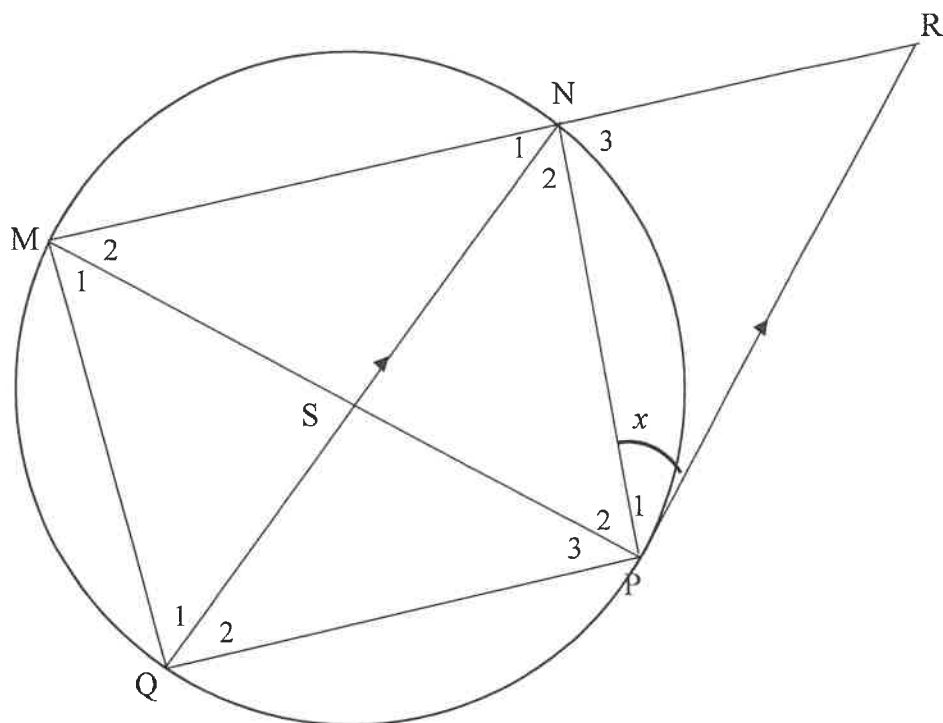
QUESTION/VRAAG 9

9.1



	Solution/Oplossing	Marks Punte
9.1		(5)

9.2

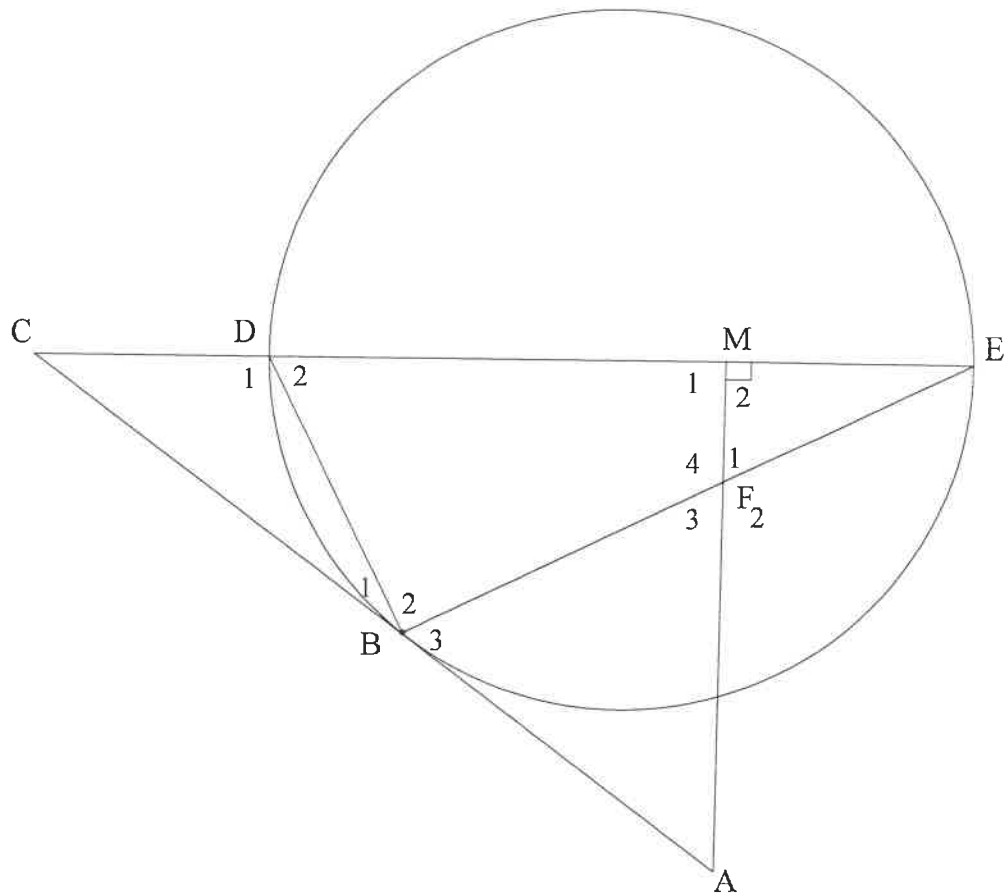


	Solution/Oplissing	Marks Punte
9.2.1(a)		(2)
9.2.1(b)		(2)
9.2.2		

	Solution/<i>Oplossing</i>	Marks <i>Punte</i>
9.2.2 (contd)		
		(6)
		[15]



QUESTION/VRAG 10



	Solution/Oplissing	Marks Punte
10.1.1		(3)
10.1.2		(4)

	Solution/Oplissing	Marks Punte
10.1.3		(3)
10.2.1		(3)
10.2.2		(4)
		[17]

[illegible]

TOTAL/TOTAAL: 150

RE-MARK/RE-CHECK HERMERK/HERSIEN				
Question <i>Vraag</i>	Marks <i>Punte</i>			Initials <i>Voorletters</i>
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
TOTAL TOTAAL				
HASH TOTAL KAF- TOTAAL				